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The Satisfaction With Life Scale and the emerging construct of life satisfaction

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Abstract

Since its introduction in 1985, the Satisfaction With Life Scale (SWLS; Diener, Emmons, Larsen, & Griffin, 1985) has been heavily used as a measure of the life satisfaction component of subjective well-being. Scores on the SWLS have been shown to correlate with measures of mental health and to be predictive of future behaviors such as suicide attempts. In the area of health psychology, the SWLS has been used to examine the subjective quality of life of people experiencing serious health concerns. At a theoretical level, extensive research conducted since the last review (Pavot & Diener, 1993) has more clearly articulated the nature of life satisfaction judgments, and the multiple forces that can exert an influence on such judgments. In this review, we examine the evolving views of life satisfaction, offer updated psychometric data for the SWLS, and discuss future issues in the assessment of life satisfaction.

Keywords: life satisfaction; subjective well-being; assessment of satisfaction; clinical assessment

Introduction

Research on 'Positive Psychology' (Seligman & Csikszentmihalyi, 2000, p. 5) has emerged as an important new direction for psychology. In sharp contrast to psychology's traditional focus on negative emotions, such as depression and anxiety, a growing number of researchers have begun to focus their efforts on the positive end of the emotional spectrum (Myers, 1992). Prominent within this growing body of research are studies focused on the experience of happiness or subjective well-being (SWB). Subjective well-being is generally conceptualized as multifaceted in nature, with both affective and cognitive components (Diener, Suh, Lucas, & Smith, 1999). Among the constituent components of SWB, life satisfaction has been identified as a distinct construct representing a cognitive and global evaluation of the quality of one's life as a whole (Pavot & Diener, 1993). Although life satisfaction is correlated with the affective components of SWB, it forms a separate factor from the other types of well-being (e.g., Lucas,

Diener, & Suh, 1996). A comprehensive assessment of SWB requires separate measures of both life satisfaction and the affective components of SWB (Diener & Seligman, 2004; Pavot, in press).

The Satisfaction With Life Scale (SWLS) was originally developed by Diener, Emmons, Larsen, and Griffin (1985), and was intended as a brief assessment of an individual's general sense of satisfaction with their life as a whole. Although the SWLS includes only five items, it has demonstrated good psychometric characteristics (Pavot & Diener, 1993). Since its introduction, the SWLS has been used in hundreds of studies to assess the life satisfaction component of SWB. The last comprehensive review of research utilizing the SWLS was published 15 years ago (Pavot & Diener, 1993), and an examination of relevant databases revealed that the vast majority of studies which have incorporated the SWLS have been completed since that time. Thus, an updated summary is warranted, along with a conceptual analysis of the processes underlying life satisfaction.

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Across the same span of time, many studies have focused on the further articulation of the life satisfaction construct without specifically using the SWLS. A number of these studies have important implications for researchers focused on life satisfaction, both at the conceptual and the methodological levels. A broader review of research on life satisfaction provides context for a more specific consideration of the SWLS.

The purposes of this review are threefold. First, we will consider research relevant to the evolving conceptualization of life satisfaction judgments, and evidence for the importance and the utility of such judgments. Second, we will present a sampling of the diverse research that has utilized the SWLS, with a focus on psychometric and normative data, as well as findings related to the areas of health and mental health. Research including cross-cultural samples and studies representing a wide range of respondents (e.g., individuals with health issues, older adults) will be emphasized. Third, we will focus on future research issues and directions.

The nature of life satisfaction

Contemporary research has assumed a three component structure for SWB: positive affect, negative affect, and life satisfaction (Andrews & Withey, 1976; Arthaud-Day, Rode, Mooney, & Near, 2005; Diener, 1984). The life satisfaction component has been conceptualized as a '... cognitive evaluation of one's life' (Diener, 1984, p. 550). Thus, life satisfaction represents an evaluative judgment. As a component of subjective well-being (SWB), life satisfaction is related to, but partially independent of, the affective aspects of SWB (Lucas et al., 1996). But the specific cognitive processes underlying the formulation of life satisfaction judgments were initially unclear. Partly because of this lack of clarity, life satisfaction judgments have been criticized with regard to their reliability and validity. Schwarz and Strack (1999), for example, presented evidence that life satisfaction judgments are based on temporarily accessible information (i.e., information from the immediate or very recent context surrounding the query) rather than representing a careful evaluation of one's life as a whole. These contextual influences, such as small fluctuations in the mood of the respondent (Schwarz & Clore, 1983), or the particular order in which items were presented on a questionnaire (Strack, Martin, & Schwarz, 1988), were found to sometimes influence responses to items assessing global SWB. If these contextual effects were proven to be large, they would reduce the reliability of self-reports of life satisfaction.

Schimmack and Oishi (2005), using a combination of meta-analysis of existing data and a set of new studies, demonstrated that the effects of substantially irrelevant factors such as item-order effects are usually relatively small compared to chronically accessible information in the formation of life satisfaction judgments. Schimmack, Diener, and Oishi (2002) presented evidence that individuals tend to rely on the same types of information to create repeated life satisfaction judgments over time, and they also found that when the sources of information used in the formulation of life satisfaction judgments do change, reported levels of life satisfaction change as well. Factors such as current mood can influence life satisfaction judgments, but in survey situations tend to produce effects that are small compared to the stable variance in the measures (Eid & Diener, 2004).

One likely source of chronically accessible information is the personality of the respondent, particularly temperament. Temperament dispositions such as extraversion and neuroticism have consistently been shown to be related to SWB (DeNeve & Cooper, 1998). Although extraversion and neuroticism have been most closely identified with the positive affect and negative affect components of SWB, respectively (Costa & McCrae, 1980), there is evidence that temperament is correlated with life satisfaction as well. Schimmack, Diener, and Oishi (2002) show that the influence of personality dispositions on life satisfaction is mediated by their influence on a person's chronic moods.

One example of evidence for top-down effects on life satisfaction comes from a large scale study conducted in the Netherlands (Stubbe, Posthuma, Boomsma, & De Geus, 2005). Data from individuals registered with the Netherlands Twin Registry (5668 participants) were analyzed and revealed that 38% of the variance of self-report responses to a Dutch translation of the SWLS was attributable to heritability (broadly-defined), whereas the remaining variance could be attributed to unique environmental factors and error of measurement. These results reinforce the view that top-down factors exert an influence on an individual's life satisfaction, but they also indicate that unique environmental (situational, contextual) effects can modify the top-down influence.

The study by Stubbe et al. (2005) points to a second important source of chronically accessible information for life satisfaction judgments: levels of satisfaction with specific life domains. In contrast to life satisfaction, domain satisfaction reflects a judgment of a specific aspect of one's life (e.g., job satisfaction, marital satisfaction, satisfaction with housing, and so forth). Generally, domain satisfaction and life satisfaction have been shown to correlate substantially.

Initially, the nature of the relationship between domain satisfaction and life satisfaction remained unclear. Some researchers followed a top-down model in which the personality or temperament of the individual was a strong determinant of life satisfaction judgments, and life satisfaction judgments in turn influenced judgments of satisfaction within various specific domains. Such models emphasize the stability of life satisfaction and de-emphasize the situation-produced changes in various domain satisfactions. Others, who subscribe to a bottom-up approach, view life satisfaction as being determined by the summation of significant and salient domain satisfactions. From the perspective of this model, situation-induced changes in domain satisfaction combine to produce variability in life satisfaction judgments over time. In a major work, Heller, Watson, and Ilies (2004) used a meta-analytic approach to test the viability of several top-down and bottom up models. Their findings indicate that extraversion and neuroticism, along with agreeableness and conscientiousness, are related to both life satisfaction and satisfaction within various specific domains. Path analyses did not support a simple top-down model, but did support two other models: a temperament-based top-down model of life satisfaction, and an integrative model which included the direct influence of domain satisfaction on life satisfaction. These results indicate that situational factors, in conjunction with temperament, influence life satisfaction judgments, and point in the direction of a need for more comprehensive explanations than either simple top-down or bottom-up models can offer.

A long-standing concept with strong implications for research on life satisfaction and SWB has seen considerable modification in recent years: the concept of adaptation. Several top-down models of SWB have used the concept of adaptation to explain the often-observed finding that personality variables account for more of the variance of SWB than is typically accounted for by life events and changes in life domains (e.g., Brickman & Campbell, 1971; Headey & Wearing, 1992). According to the principle of adaptation, life events or changes in life domains, either positive or negative in nature, may indeed initially influence one's level of life satisfaction or overall SWB, but their impact is short-lived. People soon adapt to their new circumstances, and their level of SWB returns to a level similar to that reported before the event or change occurred. The original conceptualization of the dynamic equilibrium model (Headey & Wearing, 1992), for example, proposed that the long-term, chronic baseline level of SWB is regulated by temperament or personality. According to this model, life events or changes can exert a temporary influence on an individual's level of SWB

but, over time, people tend to return to their long-term baseline level of SWB. From this perspective, the influence of ongoing life experience on SWB is minimized, and the influence of temperament is highlighted.

Although the process of adaptation does appear to have a broad, general stabilizing influence on life satisfaction, there is mounting evidence that the impact of at least some life events and domain changes may be lasting. For example, Lucas, Clark, Georgellis, and Diener (2003) found that unemployment can have a lasting impact on life satisfaction, even after an individual is re-employed. Stressful or traumatic experiences, such as acting as a caregiver for a person with Alzheimer's disease (Vitaliano, Russo, Young, Becker, & Maiuro, 1991), or becoming a widow (Lucas et al., 2003; Stroebe, Stroebe, Abakoumkin, & Schut, 1996) can produce long-term negative impacts on SWB. Using data from two nationally representative panel studies, Lucas (2007) found that acquiring a long-term disability was associated with moderate to large drops in SWB, and was followed by relatively little adaptation. In light of emerging evidence that SWB levels do sometimes change over time, Headey (2006) has proposed a modification of dynamic equilibrium theory, identifying personality traits and life events that are associated with such change. Thus, events in at least some life domains appear to have the power to bring about long-term variations in an individual's level of SWB. For a more complete discussion of contemporary research findings on adaptation, the reader is directed to Diener, Lucas, and Scollon (2006).

Based on their analysis of a large ($N=3608$) longitudinal German panel study, Fujita and Diener (2005) proposed a 'soft baseline' (p. 162) conceptualization to explain the dynamics of life satisfaction. Their study revealed modest stability in life satisfaction (in this case, over a period of 17 years), but also indicated that some individuals report substantial fluctuations, and that overall stability in life satisfaction was lower than the observed stability in physical variables and personality dispositions (Fujita & Diener, 2005). Heller, Watson, and Ilies (2006) presented evidence for substantial intra-individual variation in life satisfaction, which could be systematically linked to changes in life domains such as marital and job satisfaction.

Life satisfaction judgments appear to be based primarily on chronically accessible information. These chronically accessible sources of information include satisfaction in important life domains, as well as a person's moods and emotions, which in turn are affected by temperament. When university students were asked to report the sources of their life satisfaction judgments (Schimmack, Diener, &

Oishi, 2002), they reported thinking about life domains such as academic performance, romantic relationships, and family relationships. These domains were generally rated as more important sources of information than other potential sources, such as the weather or the performance of the university basketball team. Schimmack, Diener et al. (2002) also found individual differences in the use of sources, and in the importance assigned to particular domains. Individuals who rated a particular domain as important also tended to indicate that domain as a source of information in the formation of life satisfaction judgments. Also, this domain correlated more with life satisfaction than it did for other people who said it was not important. Individuals tend to draw in information from the same domains when forming life satisfaction judgments; when the domains used in forming life satisfaction judgments do change, reported levels of life satisfaction also often change. Thus, life satisfaction judgments do appear to be sensitive to changes in the important domains of a person's life. Schimmack and Oishi (2005) conclude that '... domain satisfaction is the most proximal determinant of life satisfaction, and examining the determinants of domain satisfaction can provide important information about the determinants of life satisfaction' (p. 404).

Perceptions of life satisfaction are likely to be influenced by personality traits, most notably extraversion and neuroticism, but changes in life domains and major life events also appear to provide salient input. The influence of immediate contextual factors such as transient mood states or item-order effects must be recognized, but these 'noise' variables generally do not eliminate the 'signal' of life satisfaction judgments. Thus, changes in life satisfaction appear to be more systematically tied to changes in chronically accessible domains, rather than being the product of random and transient contextual factors (Schimmack et al., 2002; Schimmack & Oishi, 2005). Because life satisfaction includes information from the important domains in a person's life, it provides an integrated judgment of how the person's life as a whole is going.

The importance of life satisfaction judgments

The effort that has been expended to articulate the nature of life satisfaction judgments would be of little consequence if the validity and utility of such judgments could not be established. The importance and utility of life satisfaction judgments have been questioned on at least two levels. One of these levels is specific: the level of assessment. Do reports of life satisfaction add to the valid assessment of SWB, over and above hedonically-based measures of affect?

A second question is more general: does the experience of life satisfaction and, more generally, happiness, really matter? Is SWB a valuable asset, and does it influence a person's behavior and chances for positive life outcomes?

Some have questioned the value of measures of life satisfaction in the assessment of well-being. Generally espousing the hedonistic tradition and its focus on the maximization of pleasurable experiences as the source of happiness (Diener & Fujita, 2005), these investigators tend to focus their attention on the assessment of the affective components of well-being. For example, Kahneman (1999) suggested that happiness is best conceptualized as the experience of the most pleasure over time. From this perspective, 'Objective happiness is defined by the average of utility over a period of time' (Kahneman, 1999, p. 3). Following this reasoning, the best measure of SWB would involve the repeated sampling of affective states across many specific moments. Further, the reports of affective states should be obtained as close to the moment of their experience as possible. Techniques such as Experience-Sampling Methodology (Diener, 2000; Scollon, Kim-Prieto, & Diener, 2003) represent assessment methodologies that, from the hedonistic perspective, provide optimal measurement of objective happiness. From this perspective, global assessments such as life satisfaction judgments are prone to memory errors and integration errors; the gold standard of well-being is thought to be pleasant moments.

Other scholars have adopted a more holistic approach to conceptualizing happiness. Following philosophers such as Sumner (1996), some contemporary scholars see the essence of SWB as involving reflective, global judgments of life overall (Veenhoven, 1984). From a holistic perspective, SWB includes positive affect, but it also includes other facets of life, such as engagement with life and meaning (Seligman, 2002; Peterson, Park, & Seligman, 2005). From this perspective, an accurate summation of moment-to-moment affect provides a partial assessment of overall SWB, but it does not capture all aspects of experience that are important to the person. Measures of life satisfaction are advantageous because they allow respondents to determine their own criteria for inclusion in the judgment process, and to weight them in the manner they choose. Furthermore, the information considered is not limited to affective experiences (Diener, Scollon, & Lucas, 2003), but can include nonaffective information, such as the person's success at reaching valued life goals. It is important to note that even the experiences of emotions can be weighted differently by different respondents. Because people may differentially value different positive emotions, summing emotions does not inevitably provide an accurate

assessment of how people evaluate their lives. Although life satisfaction judgments may be to some extent influenced by transient contextual factors as discussed above, most of the information used in such judgments is of a chronically accessible nature (Schimmack & Oishi, 2005), and is likely to provide a valid positive increment in a holistic assessment of SWB, over and above moment to moment assessments of affective states. Thus, this perspective is consistent with Sumner (1996) and philosophers who say the good life is when a person experiencing it *thinks it is*, after reflecting on it. Because a person may include factors in such judgments that are not salient on a moment-to-moment basis in everyday life, life satisfaction judgments will differ systematically from reports of pleasant and unpleasant affect. Feeling successful or unsuccessful in important life domains can influence life satisfaction even when these domains do not substantially influence affect.

In an examination of these two forms of satisfaction, the sum of good moments compared to a self-reflective global evaluation of life, Diener and Fujita (2005) found that respondents who reported high levels of life satisfaction, after controlling for a measure of daily satisfaction, rated themselves and were rated by informants as being higher in desirable attributes such as health, social skills, and energy, and lower in suicidal attempts and ideation. Thus, global measures, such as life satisfaction assessments, can provide important additional predictive power, over and above moment-to-moment assessments of affect. An assessment that is intended to be comprehensive should include measures of the several separable SWB concepts (Diener & Seligman, 2004). From an assessment perspective, measures of both the affective and cognitive components of SWB are important to provide the most complete measurement possible.

From a more general perspective, does it really matter if we are satisfied? That is to say, is the experience of happiness and life satisfaction important enough to justify assessment? Some scholars have argued that happiness does not really matter, whereas others have seen happiness as detrimental to other aspects of life, such as productivity (see Veenhoven, 1984).

In important life domains such as the quality of both social and marital relationships, success in work settings, physical health outcomes, and positive mental health, empirical evidence indicates that the experience of SWB and life satisfaction can be beneficial. Individuals reporting high SWB have stronger social relationships (Diener & Seligman, 2002), and marital satisfaction (Glenn & Weaver, 1981). Borrello (2005) found that college students who reported higher levels of SWB at the beginning of a semester experienced significantly greater academic success at the end of the term. Life satisfaction

is predictive of reduced suicide risk (Koivumaa-Honkanen, et al. 2001). From the perspective of physical risk, life satisfaction has been shown to be associated with reduced risk of aortic calcification in healthy women (Matthews, Owens, Edmundowicz, Lee, & Kuller, 2006). Other studies have indicated relations between various components of SWB and physical health, longevity, and survival (Lyubomirsky, King, & Diener, 2005).

When considered at the level of assessment, life satisfaction judgments can account for valid increments in explained variance, over and above measures of the affective components of SWB (Lucas et al., 1996). Life satisfaction judgments represent summary judgments that likely include additional aspects of SWB, such as evaluations relating to meaning in life, not captured by measures such as ESM. From a broader perspective, there is extensive evidence that high levels of life satisfaction and positive affect (i.e., happiness) are related to a wide range of important life outcomes, such as both physical and mental health and social relations.

The Satisfaction With Life Scale

The SWLS is intended to assess an individual's global judgment of her or his life satisfaction (Diener, Emmons, Larsen, & Griffin, 1985). The authors began the development of the SWLS by generating a pool of 48 items intended to reflect life satisfaction and well-being. From this original pool of items, factor analysis was used to identify 10 items with high loadings (0.60 or above) on a common factor interpreted as global evaluations of a person's life. After the elimination of redundancies, this group of items was then further reduced to five items, with minimal effect on the alpha reliability of the scale. A 7-point Likert style response scale (ranging from 1 = strongly disagree to 7 = strongly agree) was utilized in order to afford respondents an array of response options. The five items are all keyed in a positive direction, so the five responses can simply be added to arrive at a total score for the scale. The possible range of scores is therefore 5 to 35, with a score of 20 representing the neutral point on the scale. Scores between 5 and 9 indicate that the respondent is extremely dissatisfied with life, whereas scores ranging between 31 and 35 indicate that the respondent is extremely satisfied with life. Scores between 21 and 25 represent slightly satisfied, and scores from 15 to 19 are interpreted as falling in the slightly dissatisfied range. For a complete description of the development of the SWLS, the reader is referred to Diener et al. (1985). The SWLS was developed to be useful in the assessment of people with a wide range of ages and groups; other, more specific assessments have been

developed for particular populations, such as elementary school students (Huebner, 1994).

After the original development of the SWLS, subsequent research provided additional psychometric data on the scale. For example, Pavot and Diener (1993) present data from six studies in which the coefficient alpha for the SWLS ranged from 0.79 to 0.89, indicating that the scale has high internal consistency. More recently, Adler and Fagley (2005), and Steger, Frazier, Oishi, and Kaler (2006) reported coefficient alphas of 0.87 and 0.86, respectively, for the scale. Examples of test-retest reliability include coefficients of 0.84 (Pavot, Diener, Colvin, & Sandvik, 1991) and 0.80 (Steger et al., 2006) for a 1-month interval, and 0.54 over a 4-year span (Magnus, Diener, Fujita, & Pavot, 1993). These test-retest correlations indicate that life satisfaction has moderate temporal stability, but is also subject to change over time, reinforcing the findings of Fujita and Diener (2005), who examined the stability of life satisfaction using data from 17 years of a large representative panel study from Germany. In this sample, average life satisfaction for the first 5 years of the period correlated 0.51 with average life satisfaction for the last 5 years of the study (Fujita & Diener, 2005). A recent meta-analysis (Vassar, 2008), including reliability data for the SWLS from more than 60 studies, provides much additional information regarding scale performance and reliability across a number of different sample characteristics.

Several studies have examined the relation between the SWLS and various clinical measures of distress. Blais, Vallerand, Pelletier, and Briere (1989) reported a strong negative correlation ($r = -0.72$) between the SWLS and the Beck Depression Inventory (Beck, Ward, Mendelson, Mock, & Erbaugh, 1961). Others have replicated the depression/SWLS relation, albeit finding it to be more moderate, such as $r = -0.57$ and $r = -0.55$ over two measurement occasions (Schimmack, Oishi, Furr, & Funder, 2004). Arrindell, Meeuwesen, and Huyse (1991) found the SWLS to be negatively correlated with all of the symptom dimensions of the Symptom Checklist-90 (Derogatis, 1977). Larsen, Diener, and Emmons (1985) found the SWLS to be negatively correlated with a measure of negative affect.

Data from the SWLS have been subjected to factor analysis in a number of studies. As part of the original development of the scale, Diener et al. (1985) performed a principal-axis factor analysis on the SWLS. The analysis resulted in a single factor solution. This single factor accounted for 66% of the variance of the scale. The results of several subsequent factor analyses are reported in Pavot and Diener (1993). Generally, the single-factor solution has been replicated. In many of these analyses, the

fifth item of the scale ('If I could live my life over, I would change almost nothing') shows lower factor loadings and item-total correlations than the first four items of the scale (e.g., Senecal, Nouwen, & White, 2000). It appears that responses to this item, and its implicit reference to the past, involve a somewhat different cognitive search than the responses to the other items of the scale that imply a focus on the present. The judgments in response to the 'present' items reflect a horizontal evaluation (all aspects of my life right now), or a temporal summation (e.g., my whole life, or my whole adult life), whereas the judgment in response to the 'past' oriented fifth item more strongly implies a summary evaluation over years.

The accumulating evidence, from factor analysis and more sophisticated methodology (e.g., Oishi, 2006; further discussed in a later section) indicate that the fifth item of the SWLS is somewhat distinct from the other four items of the scale. Still, this item is highly correlated with the others, and need not be dropped from the measure. If a researcher is particularly interested in the respondent's satisfaction with his or her current life, the data can be examined without including the fifth item. Another alternative, if the temporal orientation of response is a particular concern, would be to consider using the Temporal Satisfaction With Life Scale (TSWLS; Pavot, Diener, & Suh, 1998), a 15-item measure which features the original five SWLS items, reworded with specific reference to the past, to the present, and to the future.

For some populations, normative ranges for scores on the SWLS are fairly well-established. Pavot and Diener (1993) presented average SWLS scores from five independent samples of US college students. Mean scores on the SWLS for these groups ranged from 23.0 to 25.2, with standard deviations ranging from 5.8 to 6.4. The weighted average across these samples (total $N = 1179$) was 23.9. Thus, it is typical for US college students to score in the 'slightly positive' range on the SWLS. Samples of adults in midlife (e.g., George, 1991) and older adults (e.g., Blais et al., 1989) also have yielded scores indicating levels of satisfaction well above the neutral point, with averages ranging from 23.6 to 27.9, respectively.

In contrast to these groups, other samples have indicated relatively low levels of life satisfaction. Average scores for male prison inmates (Joy, 1990) and sex workers (Baker, Wilson, & Winebarger, 2004) were 12.3 and 10.3, respectively, indicating distinctly low levels of life satisfaction. Individuals with health concerns also typically report relatively low life satisfaction, although there is considerable variation between such groups (see Table 1; see also Pavot & Diener, 1993).

Table 1. Representative data for the Satisfaction With Life Scale.

Sample characteristics		N	Mean	SD
International/cross-cultural samples				
1. English adults	Male adults	206	23.0	6.8
	Female adults	214	23.7	6.7
2. English adults		111	24.1	6.9
3. Dutch adults		1742	26.9	5.7
4. Dutch adults	Male adults	1083	25.6	5.3
	Female adults	1431	25.7	5.8
5. Australian adults		191	24.9	6.0
6. Australian late adolescents/adults		107	23.6	12.2
7. Spanish university students		161	24.4	5.6
8. National Taiwan university Students	Male students	207	20.4	7.0
	Female students	269	21.1	6.7
9. Japanese college students		78	21.0	5.5
10. Korean college students	Male students	205	20.4	4.9
	Female students	143	19.2	5.0
11. Belarusian university students		74	16.2	4.6
12. African American college students (USA)		114	22.4	6.4
13. Female college students (USA)	Black students	150	24.2	6.5
	White students	150	27.1	5.3
14. French-Canadian university students		144	25.4	5.8
15. Maasai (Southwestern Kenya)		127	27.0	3.0
16. Amish (Illinois)		52	22.0	5.0
17. Inughuit (Greenland)		179	25.0	5.0
18. Gifted adults (USA)		99	20.2	3.3
Clinical and counseling samples				
19. Psychiatric patients		474	20.1	7.8
20. Individuals with traumatic brain injury		70	19.1	3.3
	Injury and PTSD	26	12.9	5.5
21. Clinical clients (private practice)	Intake group	27	14.4	6.7
	Advanced group	16	18.3	7.1
22. Elderly caregivers	Time 1	79	21.2	7.7
	Time 2	79	19.7	8.1
23. Holocaust survivors	Group 1	160	23.3	6.2
	Group 2	87	23.0	6.0
Health-related samples				
24. Dutch medical outpatients		107	23.6	7.0
25. Lung transplant candidates (USA)		93	18.8	8.4
26. Adults with spinal cord injury (12 months post injury)		33	17.6	7.1
27. Adults with spinal cord injury	Single	53	16.1	7.6
	Married	53	19.7	7.9
28. Older adults in visual rehabilitation program		54	25.6	6.2
29. Bone marrow transplant patients (Canadian)		25	24.3	6.5
30. Individuals with diabetes (French Canadian)		638	24.9	n.r.
31. Spouses of people with fibromyalgia syndrome (USA)		135	24.3	7.3

Notes: n.r. = not reported. Some studies reported SWLS scores as the average of a single item; these scores were multiplied by five to establish a standard for comparison. Samples: 1 = Maltby & Day, 2004; 2 = Hayes & Joseph, 2003; 3 = Arrindell, Heesink, & Feij, 1999; 4 = van Loon, Tijhuis, Surtees, & Ormel, 2001; 5 = Gannon & Ranzijn, 2005; 6 = Palmer, Donaldson, & Stough, 2002; 7 = Extremera & Fernandez-Berrocá, 2005; 8 = Wu & Yao, 2006; 9 = Oishi & Sullivan, 2005; 10 = Cha, 2003; 11 = Pavot & Weinhold, 1994; 12 = Barnes & Lightsey, 2005; 13 = Chang, Watkins, & Banks, 2004; 14 = Tremblay, Blanchard, Pelletier, & Vallerand, 2006; 15, 16, 17 = Biswas-Diener, Vitterso, & Diener, 2005; 18 = Perrone, Webb, Wright, Jackson, & Ksiazak, 2006; 19 = Arrindell et al., 2001; 20 = Bryant et al., 2001; 21 = Friedman, 1991; 22 = Vitaliano et al., 1991; 23 = Shmotkin & Lomranz, 1998; 24 = Arrindell et al., 1991; 25 = Rodrigue, Kanasky, Marhefka, Perri, & Baz, 2001; 26 = Elliott et al., 2001; 27 = Putzke et al., 2001; 28 = Dreer et al., 2005; 28 = Courneya, Keats, & Turner, 2000; 30 = Senecal et al., 2000; 31 = Bigatti and Cronan, 2002.

Data for the SWLS from an array of international/cross-cultural, health-related, and counseling-related studies are presented in Table 1. The data table is not intended to be all-inclusive; it is intended to be representative of current studies that have included the SWLS, and to provide some benchmarks for

researchers focused on a particular area (e.g., individuals with health concerns or disabilities). The selected studies were not chosen in random or haphazard fashion; they were chosen in order to give readers an idea of the full range of satisfaction encountered within various groups, from extremely

Table 2. Correlations of personality and emotion variables with the Satisfaction With Life Scale (SWLS).

Variable	Sample reference number								
	1	2	3	4	5	6	7	8	9
Positive affect	0.47	0.41	0.44	0.36	0.55				
Negative affect	-0.44	-0.44	-0.57	-0.40	-0.44				
Self-esteem	0.64			0.58		0.55			
Optimism		0.48				0.50			
Pessimism		-0.45							
Depression		-0.50					-0.56	-0.57	
Extraversion								0.42	0.42
Neuroticism								-0.49	-0.54
Perceived stress		-0.56					-0.52		
Suicide ideation		-0.44							

Notes: 1 = Kang, Shaver, Sue, Min, & Jing, 2003; 2 = Chang & Sanna, 2001; 3 = Chang et al., 2004; 4 = Arrindell et al., 2001; 5 = Palmer et al., 2002; 6 = Steger et al., 2006; 7 = Tremblay et al., 2006; 8 = Schimmack et al., 2004; 9 = Hayes & Joseph, 2003.

low to very high life satisfaction. The listed studies were also intended to present some important comparison numbers for groups that are the focal point of much current research. Correlations between the SWLS and an array of personality and emotion variables are presented in Table 2.

Many additional reports, in most cases representing high quality research, were reviewed but not included in this report. We sought a balance point between a too brief report of limited scope and a completely comprehensive review of necessarily ponderous proportions. One consideration was the avoidance of redundancy. The findings of many studies of college men and women in the USA, for example, tended to yield very similar results with regard to the SWLS. Conversely, international studies of college students often revealed interesting differences, relative to comparison groups in the USA. Thus, several international samples are included. Another issue involved the way in which data from the SWLS were analyzed. A number of studies (e.g., Hindelang, Schwerin, & Farmer, 2004; King & Raspin, 2004; Malka & Chatman, 2003) used the SWLS as one measure among several intended to assess the cognitive and affective components of SWB. These multiple measures were then melded to form a composite measure of SWB, and specific results for the SWLS were minimally reported or not reported. Other researchers (e.g., Elliot, Gable, & Mapes, 2006) used the SWLS as a basis for constructing a more specific domain satisfaction scale, such as the 'Satisfaction With Social Bonds Scale.' In a few other cases, authors cited the SWLS in a discussion of life satisfaction or SWB, but they did not actually use the scale in their study. The studies that have been included represent a small but systematically selected sample of many hundreds of reports which include data from the SWLS.

Cross culturally-relevant studies

The SWLS has been used in a number of cross-cultural studies that have examined life satisfaction and, more generally, SWB. Researchers have translated the SWLS into a growing array of languages, including Spanish, German, and Japanese (Suh, Diener, Oishi, & Triandis, 1998), French (Blais et al., 1989), Chinese (Shao, 1993), Korean (Suh, 1994), Russian (Balatsky & Diener, 1993), Czech (Lewis, Shevlin, Smekal, & Dorahy, 1999), and Arabic (Abdallah, 1998). Early cross-cultural research on SWB and life satisfaction tended to focus on differing levels of SWB reported by one culture versus another. Large differences in SWB were observed between nations; these differences were found to be related to factors such as national wealth and democratic governance (Inglehart, 1990).

More recently, cross-cultural research revealed both differences and similarities in the nature of life satisfaction judgments across diverse socio-cultural groups. Diener and Diener (1995) found that people in individualistic nations were likely to weight self-esteem more than collectivists in their life satisfaction judgments, and that financial satisfaction was a stronger correlate of life satisfaction in poor nations than in wealthier societies. Suh et al. (1998) examined two large international data sets, and found that at the individual level, life satisfaction judgments were strongly related to emotions in individualists cultures, whereas social norms (e.g., the normative value of life satisfaction) were equally important to emotions when predicting life satisfaction in collectivist cultures. A study by Schimmack, et al. (2002) examined data from three collectivistic cultures (Japan, Mexico, and Ghana) and two individualistic cultures (Germany and the USA) to determine the influence of culture on the relation between personality and the components of SWB. Their findings indicate that the

influence of personality on the emotional component of SWB is pancultural, but that the influence of personality on life satisfaction was moderated by culture. Thus, the information that enters into people's life satisfaction judgments can vary across cultures.

As more sophisticated approaches are applied to the analysis of survey data for cross-cultural comparisons, it has become apparent that, in some cases, comparisons based on raw scores on measures such as the SWLS may be misleading. As an example, Vitterso, Biswas-Diener, and Diener (2005) compared SWLS scores obtained from a sample of Norwegians with the SWLS scores from a sample of individuals in Greenland. Although the initial analyses showed no mean difference between these groups, when item-response modeling was applied to the data, it was found that the Greenlanders were more prone to random responding, and using extreme response categories. After controlling for these tendencies, Norwegians were found to be more satisfied than Greenlanders, with the exception of a specific latent class of Greenlanders, who were in turn more satisfied than the Norwegian sample. Oishi (2006) examined measurement equivalence of the SWLS with Chinese and American samples, using Structural Equation Modeling (SEM), multiple indicator multiple cause modeling (MIMIC), and Item Response Theory (IRT) analysis. He found only one biased item across cultures with the SEM and MIMIC methods, but found differential item functioning for four of the five items using IRT analysis. Further, the latent mean scores of the Chinese sample were substantially lower than those of the Americans (Oishi, 2006). Tucker, Ozer, Lyubomirsky, and Boehm (2006) also found some evidence for the lack of comparability between Russian and North American groups in terms of their self-reports of life satisfaction; their findings point to the importance of testing for measurement invariance in the SWLS. These studies serve to caution researchers who are evaluating cross-cultural survey data, and at the same time offer some new approaches for the analysis of such data, and insights into how culture influences responding to various questions.

Data from other diverse populations

Studies focused on life satisfaction and SWB have often used university students as participants, likely due to factors such as convenience and availability. Gradually, however, the SWLS has come to be used in research involving a wide range of populations. Groups as diverse as male prison inmates (Joy, 1990) and contemplative nuns (McGarrahan, 1991) have completed the SWLS. Hindelang et al. (2004) used

the SWLS as a component of a large scale, global quality of life assessment of members of the United States Marine Corps. Frieswijk, Buunk, Steverink, and Slaets (2004) used a Dutch translation of the SWLS in a study of the effects of social comparison on the life satisfaction of frail older persons in the Netherlands.

The data from many studies that include assessments of life satisfaction indicate that most people are at least slightly satisfied with their lives, but that finding is not invariant. Situations such as the experience of severe physical injury or extremely negative environmental conditions can have a strong negative impact on life satisfaction. Bryant, Marosszeky, Crooks, Baguley, and Gurka (2001) found very low levels of life satisfaction among a group of individuals who had suffered severe traumatic brain injury; life satisfaction was particularly low among a sub-group who were also experiencing posttraumatic stress disorder. Biswas-Diener and Diener (2001) interviewed people in the slums of Calcutta, India, and found their mean ratings of general life satisfaction were slightly negative overall, and somewhat more negative for some sub-groups (e.g., sex workers, pavement dwellers). When the specific domain satisfactions of the overall group were averaged, however, that average was positive, indicating that, even in very difficult environmental circumstances, some measure of satisfaction is possible. In a similar study, conducted in a large city in the Midwestern USA (Baker et al., 2004), street-level sex workers ($N = 26$) completed a set of measures relating to their health and well-being, including the SWLS. Within this group the average score on the SWLS was 10.27, with a maximum score of 19 (the neutral point of the scale is a score of 20). This group is the lowest reported average life satisfaction of any sample that we have encountered, undoubtedly due to the extremely bad life circumstances of these individuals including violence, poverty, and drug addiction.

Studies related to clinical/counseling issues

Increasing numbers of researchers in the areas of clinical and counseling psychology have begun to use measures of SWB as a complement to the more familiar assessments of distress and dysfunction in determining levels of adjustment and outcome prognosis. The idea is that interventions should not only reduce suffering, but should also enhance the quality of life, moving people upwards in terms of positive experience, as well as reducing distress. In a review focused on clinical practice, Duckworth, Steen, and Seligman (2005) discuss the usefulness of including assessment tools from positive psychology research as

a supplement to more traditional assessments when designing clinical interventions. In their words: ‘... a “build-what’s-strong” approach to therapy may usefully supplement the traditional “fix-what’s-wrong” approach’ (Duckworth et al., 2005, p. 631). The assessment of positive experiences and personal strengths, and interventions designed to enhance SWB and strengthen personal assets, provide a potential new dimension for the process of clinical practice. Recent research, such as the work focused on the relation between gratitude and SWB (Emmons & McCullough, 2003), offers potential new directions for interventions intended to enhance SWB. Thus, the connection is growing that clinical practice should not just endeavor to alleviate misery, but should also strive to build rewarding lives, and the SWLS is one tool in assessing such outcomes.

Many studies found that the SWLS is a useful instrument in the assessment of psychiatric populations. For example, Arrindell, van Nieuwenhuizen, and Luteijn (2001) found that the average score on the SWLS was significantly lower for a sample of psychiatric patients compared to the average for a non-psychiatric control group, demonstrating the discriminatory power of the SWLS. Meyer, Rumpf, Hapke, and John (2004) observed a similar difference, and further found that the comorbidity of psychiatric disorders was associated with a further reduction in reported life satisfaction. In a study with both clinical and cross-cultural implications, Wu and Wu (2008) obtained data from people with schizophrenia in several communities in Taiwan, and found evidence that the SWLS is a valid measure of life satisfaction for this group.

One potential use of measures such as the SWLS in therapy is to assess changes in SWB and life satisfaction over the course of the treatment process. An early study by P. Friedman (personal communication, November 20, 1991; reported in Pavot & Diener, 1993) indicated a significant difference in average SWLS scores between a group of clients at the beginning of therapy versus an independent sample of clients who had received therapy for one to two months. A further group of clients completed the SWLS at the beginning of therapy and again a month later, and repeated-measures analysis indicated significantly higher SWLS scores for the second occasion. These early results indicated that the SWLS was sufficiently sensitive to detect improvement over the course of therapy. Subsequent research (Friedman & Toussaint, 2006) has yielded similar results. Vitaliano, Russo, Young, Becker, and Maiuro (1991) used the SWLS to monitor changes in the life satisfaction of elderly caregivers as their spouses proceeded through the stages of primary degenerative dementia, and found significant declines in SWLS scores over the 15–18 month period of the

study. Other studies used the SWLS as a measure of the psychological well-being and adjustment of caregivers, spouses, and family members of those with serious health conditions or physical disabilities. For example, Bigatti and Cronan (2002) used the SWLS and other measures to assess the psychological well-being of the spouses of people with fibromyalgia syndrome.

Another potential use of the SWLS in conjunction with the therapy process is the assessment of risk for psychopathology and self-destructive behavior. Using a longitudinal design, Suldo and Huebner (2004) found that adolescents who reported positive life satisfaction are at less risk of acting out when confronted with stressful events. In a large-scale ($N=5032$) study of high school students, Valois, Zullis, Huebner, and Drane (2001) found life satisfaction to be inversely related to alcohol and chemical use. Other studies (Moum, 1996; Diener & Seligman, 2002; Heisel & Flett, 2004) found life satisfaction to be a significant negative predictor of suicidal ideation. Data from a longitudinal study of the Finnish Twin cohort indicated that life satisfaction was related to a lower risk of suicide 20 years later (Koivumaa-Honkanen et al., 2001), even after controlling for other risk factors such as substance-use, age, and gender.

In addition to providing alternative assessment instruments, research on SWB, and positive psychology more generally, has implications for the creation of new interventions strategies for clinicians. Duckworth et al. (2005) provide a number of examples of interventions based on the research and principles of positive psychology. Discussions of how best to pursue strength-based research and practice are ongoing (e.g., Lopez & Magyar-Moe, 2006).

Health-related studies

A large proportion of the research that has incorporated the SWLS has been focused on physical health and health outcomes. Quality of life has become a primary concern in the evaluation of both the quality and outcome of health care (Moons, Budts, & De Geest, 2006). In a review of different conceptual approaches of quality of life, Moons et al. (2006) found life satisfaction to be the most adequate and appropriate conceptualization, as it successfully addresses all of the conceptual problems they examined with regard to health-related quality of life.

Many health-related studies have utilized the SWLS as an outcome indicator for individuals undergoing medical procedures and/or rehabilitation therapy. The SWLS has been used in studies examining the outcomes of people with traumatic brain injury (Bryant et al., 2001; Whiteneck,

Gerbart, & Cusick, 2004), diabetes mellitus (Elliot, Shewchuk, Miller, & Richards, 2001), and the adjustment of people participating in low vision rehabilitation (Dreer, Elliott, Fletcher, & Swanson, 2005). Elavsky and McAuley (2005) used the SWLS as a quality of life indicator in their investigation of the effect of physical activity on menopausal symptoms. In their study, physical activity was related to life satisfaction through the mediation of physical self-worth. In another study of the effects of chronic illness on the quality of life, Heckman (2003) used the SWLS (modified with a 4-point response scale) to assess life satisfaction among people living with HIV disease, and found gender and race differences regarding barriers to health care and perceptions of AIDS-related discrimination, and evidence of a significant effect of social support on life satisfaction.

Several studies examined various aspects of adjustment to spinal cord injury, using the SWLS among other assessments of quality of life. Putzke, Elliott, and Richards (2001) found that overall self-reported quality of life was significantly lower for single versus married individuals 12 months post-injury. Barrett, Putzke, and Richards (2002) compared elderly individuals with late (60 years of age or more) versus early (prior to age 60) onset of spinal cord injury, and found that, although quality of life assessments were similar, overall self-reported handicap was significantly greater among individuals with a late onset injury. Sherman, Devinney, and Sperling (2004) included the SWLS in a study of the effectiveness of past peer-mentoring experiences and current live-in partner on adjustment after spinal cord injury, and found that past peer mentoring experiences were associated with greater life satisfaction, whereas having a current live-in partner was associated with greater mobility and economic self-sufficiency. Elliot, Uswatte, Lewis, and Palmatier (2000), using the SWLS as an outcome measure, found that greater goal instability assessed at the onset of disability was predictive of lower satisfaction with life a year later. Their findings indicate that a flexible but durable goal orientation is correlated with optimal psychological adjustment of people with a physical disability. Thus, the SWLS has been shown to be an effective outcome measure for health related quality of life research. It is an important complement to objective health assessments, and has been demonstrated to have sufficient sensitivity to change to be a useful indicator for the effectiveness of interventions.

Future research issues/directions

In reference to the assessment of life satisfaction, several issues remain to be addressed by future

research efforts. One concern is the need for further research on the cognitive processes involved in the formulation of the life satisfaction judgment. Although much better understood today than at the time the SWLS was originally developed, several aspects of the process remain unclear. For example, it is unclear to what degree such judgments represent a vertical evaluation (all aspects of my life right now) versus a temporal summation (across my whole life), and what factor might direct the information search in one direction or another.

A more general concern for SWB research is the need to further explore the process of adaptation (Diener et al., 2006). Although we now know that adaptation is not a universal process as once was theorized, it is still an often powerful and, in many ways, poorly understood influence on SWB. It appears that adaptation may be more influential on some life domains (e.g., income) than on others. It is also unclear whether there are meaningful individual differences in the strength of the adaptation process.

An important future area for research on life satisfaction involves predictive studies. Investigators have only recently begun to focus on outcomes that are predicted by life satisfaction. Future research needs to focus not only on the factors that contribute to the experience of life satisfaction, but also on the effects life satisfaction has on future behavior and life outcomes. The indications from early research point to a wide range of benefits that are predicted by earlier levels of SWB (e.g., Lyubomirsky et al., 2005). But there is a great need for further exploration and articulation of these long-term outcomes.

Discussion

Although it is difficult to completely capture the breadth and depth of the findings from the many studies that have incorporated the SWLS, some key findings seem noteworthy. Many of these findings have relevance for the understanding of life satisfaction at the construct level. One long-standing concern regarding the construct of life satisfaction has centered on the factors that influence life satisfaction judgments. Are top-down, ubiquitous influences such as personality dispositions the source of such judgments, or are they the product of a summation of a set of bottom-up factors, such as current mood and one's immediate life circumstances?

The converging evidence available to date suggests that life satisfaction judgments represent a complex combination and summation of both top-down and bottom-up factors. Broad personality traits, such as extraversion and neuroticism, tend to 'set the tone' for subjective experiences such as life satisfaction.

Somewhat narrower facets of the personality, such as personal strivings (Emmons, 1991), also exert an influence. It is becoming increasingly clear, however, that bottom-up, circumstance factors such as satisfaction with specific life domains can exert a powerful influence on judgments of life satisfaction. Although these domains tend to be habitually accessed, there is evidence that the set of domains that are accessed when forming life satisfaction judgments can change over time, and that the relative salience of any particular domain is variable as well. Further evidence suggests that there are both individual differences and cultural norms that exert influence on the formation of life satisfaction judgments. Thus, the influences are idiosyncratic. To be sure, mental illness or health can influence the SWLS score of virtually anyone. However, many factors, even internal factors such as feelings of self-efficacy and self-esteem, vary in their importance across cultures and across the context of life.

Given the variability in influences, we might not discover the true relation between PA and life satisfaction, and NA and life satisfaction. It is likely to depend on the variation within the samples, and how they weight and value moods and emotions (internal feelings) versus external factors and achievements. Thus, for any individual person, a number of personality variables and many situational or life circumstance variables influence her or his judgment of life satisfaction. This is encouraging, as it indicates that the life satisfaction judgment is very broadly based. This finding has implications for using the SWLS to assess change, such as assessing the impact of a clinical intervention. Because the responses to the SWLS are so broadly based, it may not be extremely sensitive to interventions, unless they are large. Thus, it should often be supplemented by other measures that are focused on improvement in the particular domain of the intervention. If life satisfaction does change, it is very impressive. But there could be more focal improvements that are too subtle to influence something as broad as life satisfaction.

Another area of ongoing discussion has focused on the temporal stability of life satisfaction judgments. Early criticism of the construct of life satisfaction centered on the potential influences of current mood on life satisfaction judgments. Because current mood (one's mood at the time of completing a questionnaire) was demonstrated as a potential source of influence on life satisfaction judgments (Schwarz & Clore, 1983), it was assumed that life satisfaction judgments were based solely on current mood and immediate contextual information (e.g., a sunny versus cloudy day). More recent work (Eid & Diener, 2004) has shown that while current mood

occasionally has an effect, it is not strong, and often exerts only a small influence on judgments of life satisfaction. A more accurate depiction of the temporal influences on life satisfaction judgments incorporates a broader view of the temporal spectrum. Moment to moment changes in mood or the immediate context can occasionally influence life satisfaction judgments. Longer term and larger contextual changes (e.g., long-term unemployment, widowhood) produce larger and more lasting shifts in the judgment process. Very long-term, relatively stable influences (e.g., personality dispositions such as extraversion and neuroticism, cultural factors) tend to account for the slowest change in life satisfaction judgments.

In the same way that the factors that influence life satisfaction judgments must be understood from a multiple-level perspective, the temporal influences on life satisfaction judgments represent a complex amalgam of immediate, intermediate, and long-term components. Such a structure allows for both long-term stability and also some shorter-term fluctuation in judgments of life satisfaction. The SWLS has proven to be a reliable and valid measure of the life satisfaction component of SWB. It has shown itself to be useful in a wide range of research settings and applications. Its simple structure lends itself to easy translation into different languages, and its brevity is desirable when it is incorporated into a larger battery of assessment instruments. The empirical findings reviewed here have helped to establish the validity of the construct of life satisfaction, reveal the complexity of life satisfaction judgments, and establish the importance of life satisfaction and SWB in the experience of positive life outcomes.

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